

CodeSkyline: A Juxtaposed Visual Analytics Code-Map Embedded in a Code Editor

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Motivation

- Whenever changes occur or developers are introduced to new projects, they have to adjust their mental model to the source code.
- Many existing code-map approaches [1,2] exist outside the workflow of developers and do not scale to large projects.

Research question

- How can source code visualization with a direct mapping scale to large projects, and include high-level abstractions?
- How can a code-map metaphor be embedded into an code editor to support navigation, code comprehension, and maintenance tasks?

Approach

- We developed CodeSkyline, an VS Code extension that juxtaposes a code-map metaphor with columns to three aligned visualizations in order to gain insight into higher-abstractions, produce stable layouts across edits, and without context switching.



The visualization integrated into the VS Code Editor. The folder hierarchy is shown in a juxtaposed dendrogram. The code-map visualization shows source code files with syntax highlighting. The user searches for "timeout" and is inspecting the source code of the a file.

A juxtaposed arc visualization above the code-map. It displays file couplings, that is whether a source code file imports from another file. With these arcs, users can identify clusters of files, such as independent sub-projects or important files.



A juxtaposed bar chart showing the centrality value calculated with the PageRank algorithm [3] based on the file couplings. It enables users to find important source code files in a repository.

Rendering text on different granularities

Trades readability for contrast and comparability.



Limitations and Future Work

- The alignment increases whitespace compared to space-filling layouts. Is this tradeoff justified?
- We intend to conduct a summative qualitative evaluation of the tool to evaluate the usability of our prototype.
- Later we will investigate tool adoption in real-world contexts.

Try our prototype:



<https://codeskyline.iva.igd.fraunhofer.de/>

[1] I. Bacher, B. Mac Namee, and J. Kelleher. The Code-Map Metaphor - A Review of Its Use Within Software Visualisations. In Proceedings of the 12th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications, pp. 17–28. SCITEPRESS - Science and Technology Publications, 2017. doi: 10.5220/0006072300170028

[2] S. Eick, J. Steffen, and E. Sumner. Seesoft - a tool for visualizing line oriented software statistics. IEEE Transactions on Software Engineering, 18(11):957–968, 1992. doi: 10.1109/32.177365

[3] Brin S, Page L. The anatomy of a large-scale hypertextual web search engine. Computer networks and ISDN systems. 1998 Apr 1;30(1-7):107-17. doi: 10.1016/S0169-7552(98)00110-X.